



HK IMO Training 2019-2020
Problem Set 3 Suggested Solutions



Problem 9. Let ABC be a triangle with $\angle BAC > 90^\circ$. Let D be a point on the segment BC and E be a point on the line AD such that AB is tangent to the circumcircle of $\triangle ACD$ at A and BE is perpendicular to AD . Given that $CA = CD$ and $AE = CE$, determine $\angle BCA$.

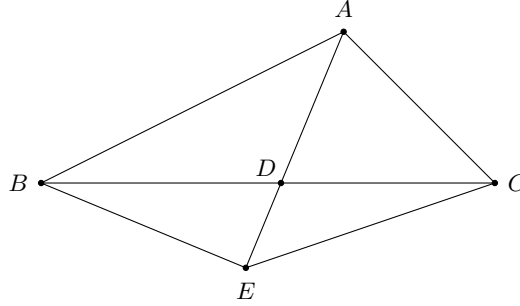
Solution 1. Let O be the circumcentre of $\triangle ABC$. Then

$$\angle EBA = 90^\circ - \angle BAE = 90^\circ - \angle ACB = \angle OBA.$$

As E and O lie on the same side of the line AB , this implies B, O, E are collinear. Also, both O and E lie on the perpendicular bisector of AC , while B does not. Therefore, O and E must coincide. It follows that

$$\angle ACB = \frac{1}{2}\angle AOB = \frac{1}{2}\angle AEB = 45^\circ.$$

Remark: The condition $CA = CD$ is redundant.



Solution 2. Let $\angle C = 2\alpha$. Then $\angle CAD = \angle CDA = 90^\circ - \alpha$. Moreover $\angle BAD = 2\alpha$ as BA is tangent to the circumcircle of $\triangle CAD$. Since $AE = CE$, it gives $\angle AEC = 2\alpha$. Thus $\triangle AEC \sim \triangle ACD$. Hence

$$\frac{AE}{AC} = \frac{AC}{AD}.$$

But the condition that $BE \perp AD$ gives $AE = AB \cos 2\alpha = c \cos 2\alpha$. It is easy to see that $\angle B = 90^\circ - 3\alpha$. Using the sine law in $\triangle ADC$, we get

$$\frac{AD}{\sin 2\alpha} = \frac{AC}{\sin (90^\circ - \alpha)}.$$

This gives $AD = 2b \sin \alpha$. Thus we get

$$b^2 = AC^2 = AE \cdot AD = (c \cos 2\alpha) \cdot 2b \sin \alpha.$$

Using $b = 2R \sin B$ and $c = 2R \sin C$, this leads to

$$\cos 3\alpha = 2 \sin 2\alpha \cos 2\alpha \sin \alpha = \sin 4\alpha \sin \alpha.$$

Writing $\cos 3\alpha = \cos (4\alpha - \alpha)$ and expanding, we get $\cos 4\alpha \cos \alpha = 0$. Therefore $\alpha = 90^\circ$ or $4\alpha = 90^\circ$. But $\alpha = 90^\circ$ is not possible as $\angle C = 2\alpha$. Therefore $4\alpha = 90^\circ$, which gives $\angle C = 2\alpha = 45^\circ$.

Problem 10. Find all positive integers n such that

$$\frac{n^{3n-2} - 3n + 1}{3n - 2}$$

is an integer.

Solution. Since $3n - 2 \mid n^{3n-2} - 3n + 1$, it follows that $3n - 2 \mid n^{3n-2} - 1$, hence $n^{3n-2} \equiv 1 \pmod{3n - 2}$.

Note that n must be odd (otherwise the even $3n - 2$ would divide the odd $n^{3n-2} - 3n + 1$, which is impossible).

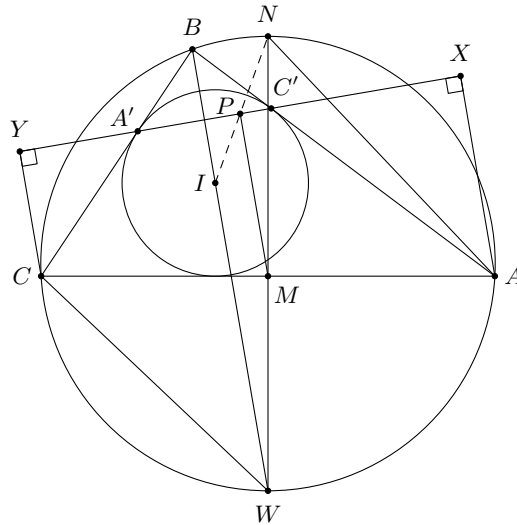
Let $p > 2$ be the smallest prime factor of the odd $3n - 2$. Obviously $p \nmid n$, hence by Fermat's little theorem we have $n^{p-1} \equiv 1 \pmod{p}$. We also have $n^{3n-2} \equiv 1 \pmod{p}$.

Let r be the order of n modulo p . Then $r \mid p - 1$ and $r \mid 3n - 2$. Since $r \mid \gcd(p - 1, 3n - 2)$ and $\gcd(p - 1, 3n - 2) = 1$, it must be $r = 1$, i.e. $n \equiv 1 \pmod{p}$.

Considering $p \mid n - 1$ and $p \mid 3n - 2$, it follows that $p \mid 1$, which contradicts the assumption about the existence of prime factor of $3n - 2$. Hence $3n - 2 = 1$, i.e. $n = 1$, and that is indeed the only solution.

Problem 11. The incircle of an acute-angled triangle ABC with centre I touches sides BC and AB at points A' and C' respectively. Let M be the midpoint of side AC , and let N be the midpoint of arc AC , containing B , in the circumcircle of $\triangle ABC$. If P is the projection of the point M to the segment $A'C'$, prove that the points I , P , and N are collinear.

Solution.



Let's denote the second intersection point of the line BI with the circumcircle of $\triangle ABC$ as W .

Clearly, W is the midpoint of the smaller arc AC , and points M , W , and N lie on the same line. It is clear that $A'C' \perp BI$, implying $PM \parallel BW$ and $\angle PMN = \angle BWN$.

Hence, it suffices to prove that $\triangle PMN \sim \triangle IWN$ (which will imply that $\angle PNM = \angle INW$ and the desired collinearity). In order to prove the similarity, we are going to show that $\frac{NM}{NW} = \frac{PM}{IW}$.

Let $\angle B = 2\beta$. Then $\angle MNA = \beta$ and

$$\frac{NM}{NW} = \frac{NM}{NA} \cdot \frac{NA}{NW} = \cos^2 \beta.$$

Let X and Y be the projections of A and C onto the line $A'C'$ respectively. Then PM is the midline of the trapezoid $AXYC$. Then we get

$$2PM = CY + XA = CA' \sin \angle YA'C + C'A \sin \angle XC'A = (CA' + AC') \cos \beta = AC \cos \beta$$

because $\angle YA'C = \angle XC'A = 90^\circ - \beta$.

Finally, it is well-known that $IW = CW$, meaning that

$$\frac{PM}{IW} = \frac{AC}{2CW} \cos \beta = \frac{CM}{CW} \cos \beta = \cos^2 \beta.$$

So we are done.

Problem 12. In a country, there are 2018 cities, some of which are connected by roads. Each city is connected to at least three other cities. It is possible to travel from any city to any other city using one or more roads. For each pair of cities, consider the shortest route between these two cities. What is the greatest number of roads that can be on such a shortest route?

Solution. The greatest number of roads that can be in such a shortest route is 1511. We first describe a country for which this number is attained. Divide the cities in 504 groups: two groups of five cities $(A_0, B_0, C_0, D_0, E_0)$ and $(A_{503}, B_{503}, C_{503}, D_{503}, E_{503})$, and groups of four cities (A_i, B_i, C_i, D_i) for $1 \leq i \leq 502$. For all i with $0 \leq i \leq 503$: connect A_i to B_i , connect A_i to C_i , connect B_i to D_i , and connect C_i to D_i . Moreover, connect E_0 to A_0 , B_0 and C_0 . Connect E_{503} to B_{503} , C_{503} , and D_{503} . For each $1 \leq i \leq 502$: connect B_i to C_i . Finally, for each $0 \leq i \leq 502$: connect D_i to A_{i+1} . Now every city is connected to exactly three other cities.

If we now want to travel from A_0 to D_{503} , then we have to travel through all A_i and D_i , because the only connections between the different groups are there, and each group is only connected to the previous and the next group. Moreover, for $0 \leq i \leq 503$, the city A_i is not connected to D_i , which causes the route within group i to go through either B_i or C_i . At least three of the four or five cities of each group must lie on the route. In total, this route visits at least $3 \cdot 504 = 1512$ cities and has at least 1511 roads.

We will now prove that the shortest route between two cities cannot have more than 1511 roads. Consider such a shortest route which visits the cities $A_0, A_1, \dots, A_k$ consecutively. Each of the cities A_i has one more neighbour besides A_{i-1} and A_{i+1} , which we will call B_i (note that the cities $B_0, B_1, \dots, B_k$ do not have to be distinct). Moreover, A_0 and A_k are connected to a third city, say $C_0 \neq B_0$ and $C_k \neq B_k$ respectively. If one of the cities B_i or C_i equals one of the cities A_j , then we could have found a shorter route by going directly from A_i to A_j (or vice versa), which would be a contradiction. Hence, the cities B_i and C_i are not equal to any of the cities A_j .

If one of the cities B_i is connected to four cities A_j , say B_i is connected to A_i, A_m, A_n , and A_p with $i < m < n < p$, then we could shorten the route by going from A_i to B_i and then to A_p . This makes us skip at least two cities of the original route (A_m and A_n) and in their place we only visit B_i , hence this new route is shorter, which would be a contradiction. Therefore, within the cities B_i with $3 \leq i \leq k-3$ there are at

least $\frac{k-5}{3}$ distinct cities. For B_0 and C_0 , the route can only be shortened if one of these cities equals B_i for certain $i \geq 3$. That would be a contradiction, hence B_0 and C_0 are two distinct cities, not equal to B_i for $i \geq 3$. In the same way, B_k and C_k are two distinct cities, not equal to B_i for $i \leq k-3$. Altogether, we have $k+1$ cities on the route itself and at least $\frac{k-5}{3} + 2 + 2$ other cities. Therefore $\frac{k-5}{3} + k + 5 \leq 2018$, hence $4k - 5 + 15 \leq 3 \cdot 2018 = 6054$, hence $4k \leq 6044$, hence $k \leq 1511$.

We conclude that the greatest number of roads occurring in a shortest route is 1511.